Notes on Multivariable Calculus

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September 25, 2025

1 Vectors

1.1 Basics

Let u, v be vectors in $\mathbb{R}^n$. They will have n components, and let them be as follows, where $u_i, v_i \in \mathbb{R}$ (scalars):

$$u = (u_1, u_2, \ldots, u_n)$$
$$v = (v_1, v_2, \ldots, v_n)$$

We can define the addition operation and the magnitude of each vector as:

$$u + v = (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n)$$
$$\|u\| = \sqrt{u_1^2 + u_2^2 + \ldots + u_n^2} = \sqrt{\sum_{i=1}^{n} u_i}$$

Since scalar addition is commutative, so is vector addition. Also, the subtraction operation is defined as:

$$u - v = u + (-v) = (u_1 - v_1, u_2 - v_2, \ldots, u_n - v_n)$$

If we have two points A and B, the vector AB from A to B is given by $AB = B - A$. If we placed the tail of the vector at A, then the head will point to B.

Two vectors are parallel if and only if they are scalar multiples of each other; that is, there exists a scalar $a \in \mathbb{R}$ such that $u = av$.

1.2 Polar Form

For a vector $u \in \mathbb{R}^2$, we can write u in polar form. Any vector can be specified with an angle from the x axis and a magnitude. Let the angle from the x axis be θ and the magnitude, the distance to the origin, be r. To convert from Cartesian to polar:

$$u = \begin{bmatrix} u_x \\ u_y \end{bmatrix} \implies \begin{cases} \theta = \arctan\left(\dfrac{u_y}{u_x}\right) \\ r = \|u\| = \sqrt{u_x^2 + u_y^2} \end{cases}$$

To convert from polar to Cartesian:

$$u = \begin{bmatrix} r\cos\theta \\ r\sin\theta \end{bmatrix} = r \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

Note that since $\sin^2\theta + \cos^2\theta = 1$ by the Pythagorean identity, the magnitude of u is r.

1.3 Unit Vectors

A unit vector is a vector of magnitude 1. For a vector $u \in \mathbb{R}^n$, a unit vector with the same direction is $u/\|u\|$.

1.4 Dot Product

For vectors $u = (u_1, \ldots, u_n) \in \mathbb{R}^n$ and $v = (v_1, \ldots, v_n) \in \mathbb{R}^n$, the cross product of the two vectors $u \cdot v$ is:

$$u \cdot v = u_1 v_1 + u_2 v_2 + \ldots + u_n v_n = \sum_{i=1}^{n} u_i v_i$$

Since scalar multiplication is commutative, so is the dot product: $u \cdot v = v \cdot u$.

Alternatively, let θ be the angle between vectors u and v. Then:

$$u \cdot v = \|u\| \|v\| \cos\theta$$

1.5 Orthogonality

Two vectors u and v are orthogonal if and only if $u \cdot v = 0$.

1.6 Standard Bases

The standard basis vectors are as follows:

$$\hat{i} = (1, 0, 0)$$
$$\hat{j} = (0, 1, 0)$$
$$\hat{k} = (0, 0, 1)$$

These vectors span the $\mathbb{R}^3$ space.

1.7 Angle Between Vectors

We can calculate the angle between vectors $u, v \in \mathbb{R}^n$ using the cross product:

$$u \cdot v = \|u\| \|v\| \cos\theta \implies \cos\theta = \frac{u \cdot v}{\|u\| \|v\|}$$

1.8 Cross Product

The cross product only exists between vectors in three dimensions. Let $u = (u_x, u_y, u_z) \in \mathbb{R}^3$ and $v = (v_x, v_y, v_z) \in \mathbb{R}^3$. The cross product $u \times v$ has the following properties:

- $u \times v$ is a vector in $\mathbb{R}^3$.
- $u \times v$ is orthogonal to both u and v. If u and v are linearly independent and span a plane, $u \times v$ is orthogonal to that plane as well.
- $u \times v = 0$ if and only if u and v are parallel/linearly dependent.
- The magnitude of $u \times v$ equals the area of the parallelogram formed by u and v: $\|u \times v\| = \|u\| \|v\| \sin\theta$

The formula for $u \times v$ can be written concisely with a determinant, and expanding by cofactors:

$$u \times v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix} = \hat{i} \begin{vmatrix} u_y & u_z \\ v_y & v_z \end{vmatrix} - \hat{j} \begin{vmatrix} u_x & u_z \\ v_x & v_z \end{vmatrix} + \hat{k} \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$$

The operation has the following properties:

- Anticommutativity: $u \times v = -(v \times u)$
- Distributive over addition: $(u + v) \times w = u \times w + v \times w$ and $u \times (v + w) = u \times v + u \times w$.
- Multiplication by a scalar: $(cu) \times v = c(u \times v) = u \times (cv)$ where $c \in \mathbb{R}$.

The scalar triple product is the determinant of the square matrix with those vectors as its rows (or columns):

$$u \cdot (v \times w) = (v \times w) \cdot u = \begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix}$$

From the determinant, it follows that the scalar triple product of u, v, w is the volume of the parallelepiped with u, v, w as its edges.

1.9 Projection

The orthogonal projection of u onto v, written $\text{proj}_v u$, is the vector parallel to v such that the error between the projection and u is minimized.

Let p be the orthogonal projection $\text{proj}_v u$. The error of the project is $e = u - p$. The error is minimized when e is orthogonal to v. The formula for orthogonal projection is:

$$\text{proj}_v u = \frac{u \cdot v}{v \cdot v} v = \frac{u \cdot v}{\|v\|^2} v$$

The proof is left as an exercise to the reader.

2 Lines

A line in 3D space can be represented in three ways. In each subsection below, we specify a line passing through (x_0, y_0, z_0) with slopes a, b, c.

2.1 Parametric

This parametric system with parameter t describes the line we want:

$$\begin{cases} x(t) = at + x_0 \\ y(t) = bt + y_0 \\ z(t) = ct + z_0 \end{cases}$$

2.2 Vector

A vector valued function $f : \mathbb{R} \to \mathbb{R}^n$. It takes a scalar and returns a vector: $f(t) \in \mathbb{R}^n, t \in \mathbb{R}$.

From the parametric equation of a line, we notice that for parameter t, this point lies on the line. We can define a vector valued function r with parameter t. This is the vector equation for a line.

$$r(t) = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} at + x_0 \\ bt + y_0 \\ ct + z_0 \end{bmatrix}$$

2.3 Symmetric

Recall the parametric system for a line. We solve for the parameter t in terms of x, y, z values:

$$\begin{cases} x(t) = at + x_0 \\ y(t) = bt + y_0 \\ z(t) = ct + z_0 \end{cases} \implies \begin{cases} t = \dfrac{x - x_0}{a} \\ t = \dfrac{y - y_0}{b} \\ t = \dfrac{z - z_0}{c} \end{cases}$$

For each equation in the above system, the parameter t must be equal, yielding the symmetric equation of a line:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

2.4 Finding Intersection

Suppose we have two lines with the following parametric equations:

$$\begin{cases} x_1(t_1) = a_1 t_1 + x_{10} \\ y_1(t_1) = b_1 t_1 + y_{10} \\ z_1(t_1) = c_1 t_1 + z_{10} \end{cases} \qquad \begin{cases} x_2(t_2) = a_2 t_2 + x_{20} \\ y_2(t_2) = b_2 t_2 + y_{20} \\ z_2(t_2) = c_2 t_2 + z_{20} \end{cases}$$

These two lines intersect exactly where $(x_1, y_1, z_1) = (x_2, y_2, z_2)$.

$$\begin{cases} a_1 t_1 + x_{10} = a_2 t_2 + x_{20} \\ b_1 t_1 + y_{10} = b_2 t_2 + y_{20} \\ c_1 t_1 + z_{10} = c_2 t_2 + z_{20} \end{cases}$$

We know the value of a, b, c, x_0, y_0, z_0 for both equations. There are two variables to solve: t_1 and t_2. But this is an overdetermined system, with three linear equations for two variables, so there is only a solution (and intersection) if these equations are not independent. If there is, then we can use t_1 and t_2 to find the point (or infinity points) of intersection.

3 Planes

The equation of a plane in 3D passing through (x_0, y_0, z_0) is:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$\begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

Alternative, expanding the first equation, there exists a scalar value d such that the plane is described by:

$$ax + by + cz + d = 0$$

Subtracting by x_0, y_0, z_0 shifts the plane. The plane parallel to the original plane, but which passes through the origin, is given by $(x, y, z) \cdot (a, b, c) = 0$. Which means that for all points on the plane, (a, b, c) is orthogonal to it. Hence $n = (a, b, c)$ is orthogonal to the plane.

3.1 Dihedral Angle

The dihedral angle is the angle formed between two planes. It equals the angle formed between the normal vectors of each plane, which can be found using the dot product.

3.2 Angle between Vector and Plane

Let v be the vector. Find the normal vector to the plane n, and find the angle θ between the vector and n. The angle between the plane and v is $90^\circ - \theta = \pi/2 - \theta$.

3.3 Finding Intersection

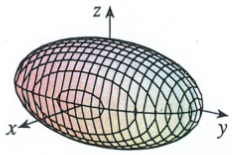
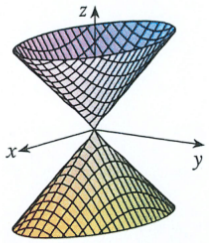
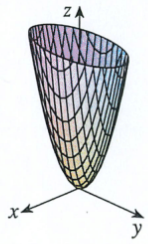
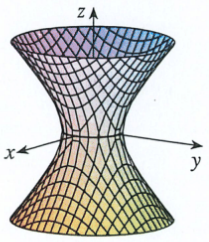
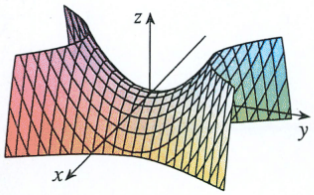
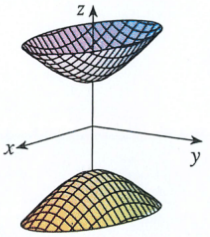
Two planes intersect if and only if the vectors normal to the plane point in different direction and are not parallel.

$$a_1x + b_1y + c_1z + d_1 = 0$$
$$a_2x + b_2y + c_2z + d_2 = 0$$

A point on the intersection of these two planes satisfy both equations. From here, set any variables (x, y, z) to a parameter t and substitute. Then use elimination to solve for the two remaining variables in terms of t (DO NOT ELIMINATE t). Now we have a parametric system describing the line of intersection.

4 Cylinders and Quadric Surfaces

Table 1 Graphs of Quadric Surfaces

Surface	Equation	Surface	Equation
Ellipsoid 	$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ All traces are ellipses. If $a=b=c$, the ellipsoid is a sphere.	**Cone** 	$\frac{z^2}{c^2}=\frac{x^2}{a^2}+\frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces in the planes $x=k$ and $y=k$ are hyperbolas if $k\neq 0$ but are pairs of lines if $k=0$.
Elliptic Paraboloid 	$\frac{z}{c}=\frac{x^2}{a^2}+\frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces are parabolas. The variable raised to the first power indicates the axis of the paraboloid.	**Hyperboloid of One Sheet** 	$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$ Horizontal traces are ellipses. Vertical traces are hyperbolas. The axis of symmetry corresponds to the variable whose coefficient is negative.
Hyperbolic Paraboloid 	$\frac{z}{c}=\frac{x^2}{a^2}-\frac{y^2}{b^2}$ Horizontal traces are hyperbolas. Vertical traces are parabolas. The case where $c<0$ is illustrated.	**Hyperboloid of Two Sheets** 	$-\frac{x^2}{a^2}-\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ Horizontal traces in $z=k$ are ellipses if $k>c$ or $k<-c$. Vertical traces are hyperbolas. The two minus signs indicate two sheets.

4.1 Trace

The intersection of a quadric surface with a plane normal to a standard basis vector is a trace. We can use the trace produced by a equation in each direction to determine which quadric surface the equation is.

4.2 Cylinders

A cylinder is a surface that consists of all lines that are parallel to a given line, and which passes through a plane curve. For example, the surface given by $y = x^2$ is a cylinder because the z component of each point can be anything, hence every line is parallel to $\hat{k}$.